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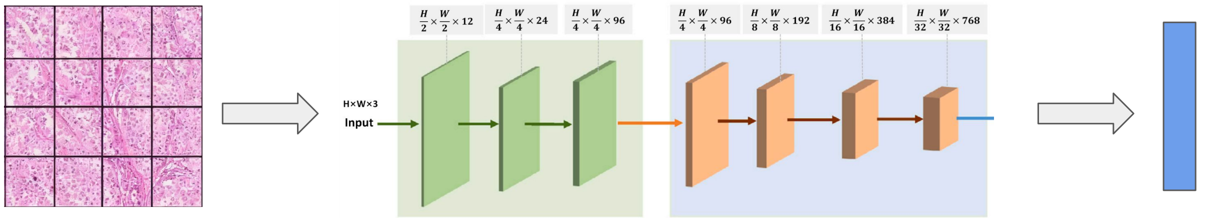
Recap

CTransPath Feature Extraction

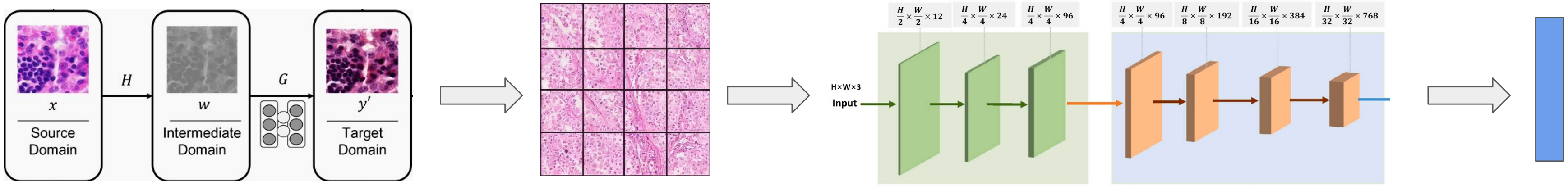


Patch Extraction

CTransPath Feature Extraction

1x768 Feature Vector

Stain Normalized CTransPath Feature Extraction



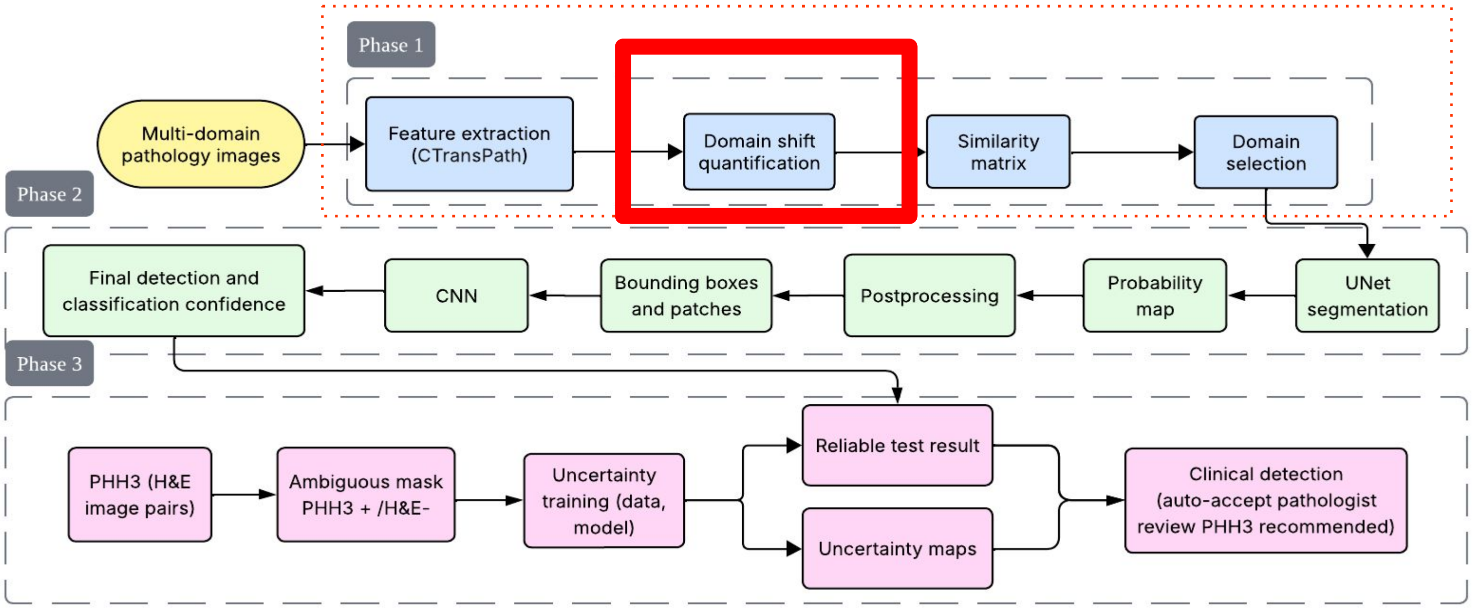
Cycle GAN Stain Normalization

Patch Extraction

CTransPath Feature Extraction

1x768 Feature Vector

Pipeline



Phase 1
Multi-domain pathology images
Feature extraction (CTransPath)
Domain shift quantification
Similarity matrix
Domain selection
Phase 2
UNet segmentation
Probability map
Postprocessing
Bounding boxes and patches
CNN
Final detection and classification confidence
Phase 3
PHH3 (H&E image pairs)
Ambiguous mask PHH3 + /H&E-
Uncertainty training (data, model)
Reliable test result
Uncertainty maps
Clinical detection (auto-accept pathologist review PHH3 recommended)

Maximum Mean Discrepancy (MMD)

Set X: 100 images of NYC

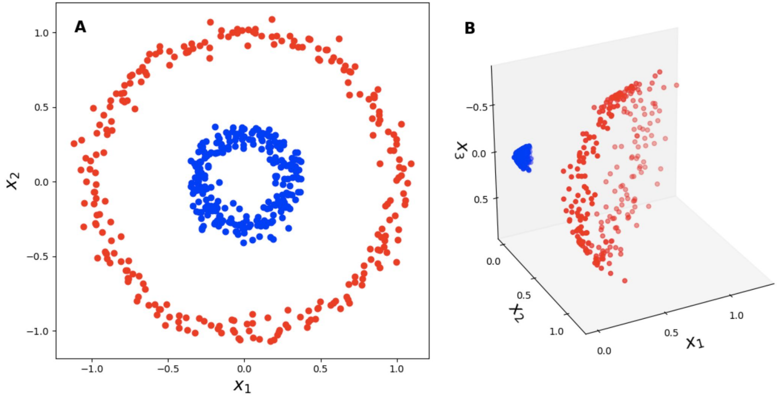
Set Y: 100 images of HK

MMD measures: How different are these collections on average?

$$\mathrm{MMD}^2 = \frac{1}{n(n-1)} \sum_i \sum_{j \neq i} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_i \sum_{j \neq i} k(y_i, y_j) - \frac{2}{nm} \sum_i \sum_j k(x_i, y_j)$$

Kernel Trick

MMD uses a kernel function to compare distributions in a high-dimensional space where differences are easier to see.



Kernel Choice: Linear Kernel (basic dot product)

$$k(x, y) = x_1y_1 + x_2y_2 + \dots + x_ny_n$$

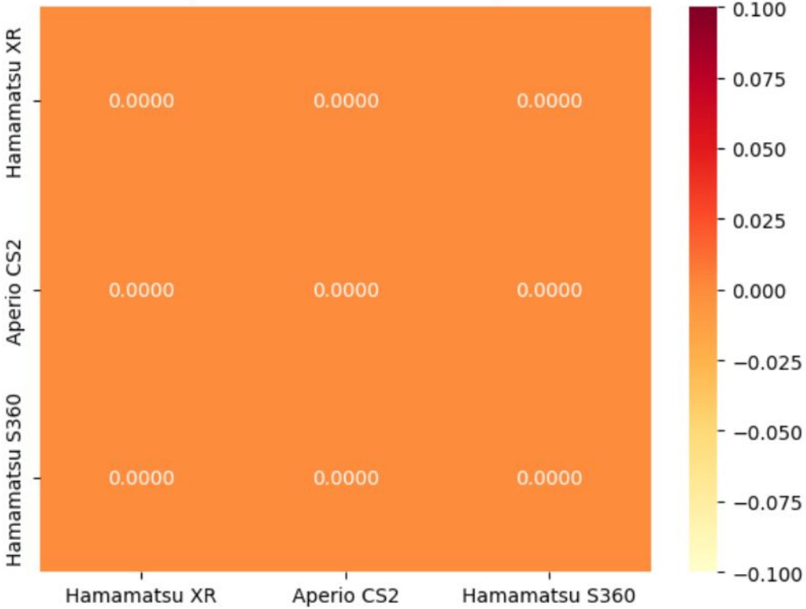
- CTransPath features already high-dimensional (768D)
- Computationally efficient - no hyperparameters (gaussian kernel requires γ)
- Interpretable - measures similarity in original space
- Stable - no gamma parameter tuning needed

Problem: Features have small means

```
Scanner Hamamatsu XR:
  Samples: (29952, 768)
  Feature mean: 0.0005
  Feature std: 0.1338
  Any NaN: False

Scanner Aperio CS2:
  Samples: (23142, 768)
  Feature mean: 0.0008
  Feature std: 0.1341
  Any NaN: False

Scanner Hamamatsu S360:
  Samples: (28015, 768)
  Feature mean: 0.0005
  Feature std: 0.1380
  Any NaN: False
```



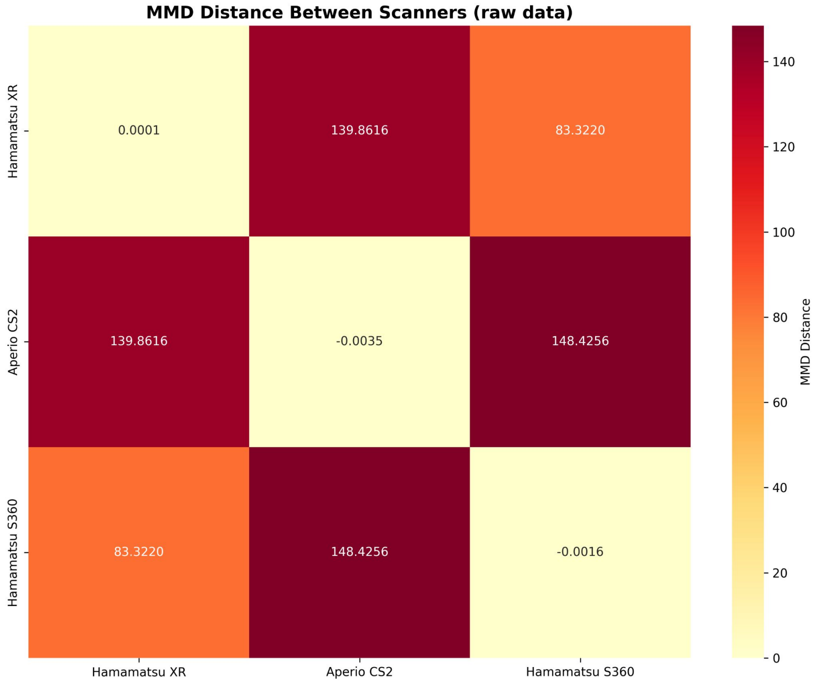
Dot product of small number ⇒ small results (floating-point precision limits)

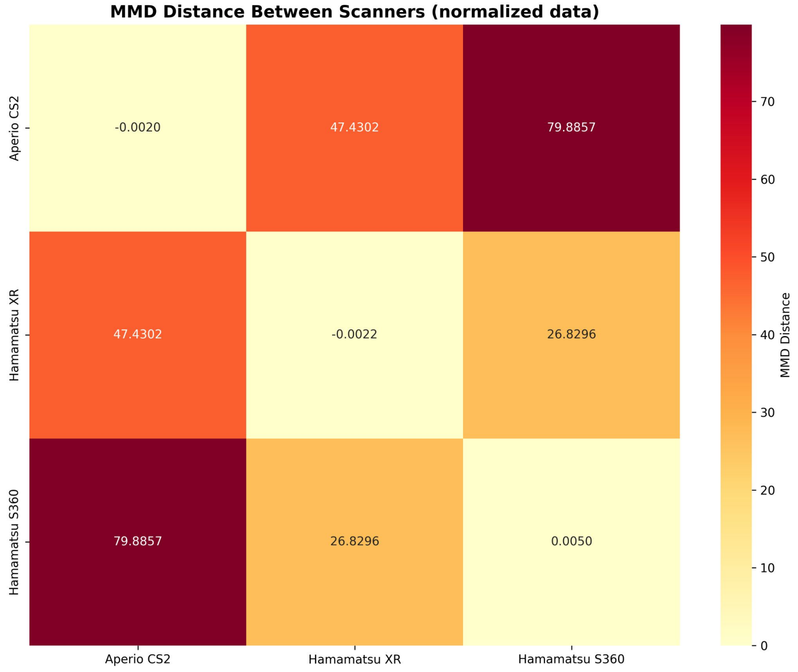
Solution: Feature Scaling (divide features by pooled standard deviation)

Scanner Differences

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23\frac{\mu m}{px}$
1b	50	breast carcinoma	UMC Utrecht	human	Hamamatsu S360 (0.5 N/A)	$0.23\frac{\mu m}{px}$
1c	50	breast carcinoma	UMC Utrecht	human	Leica ScanScope CS2	$0.25\frac{\mu m}{px}$

Scanner Differences

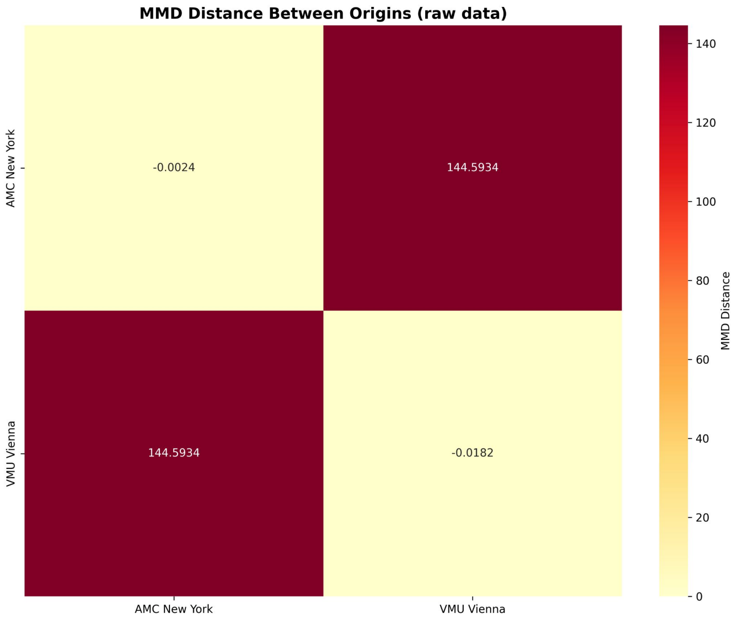


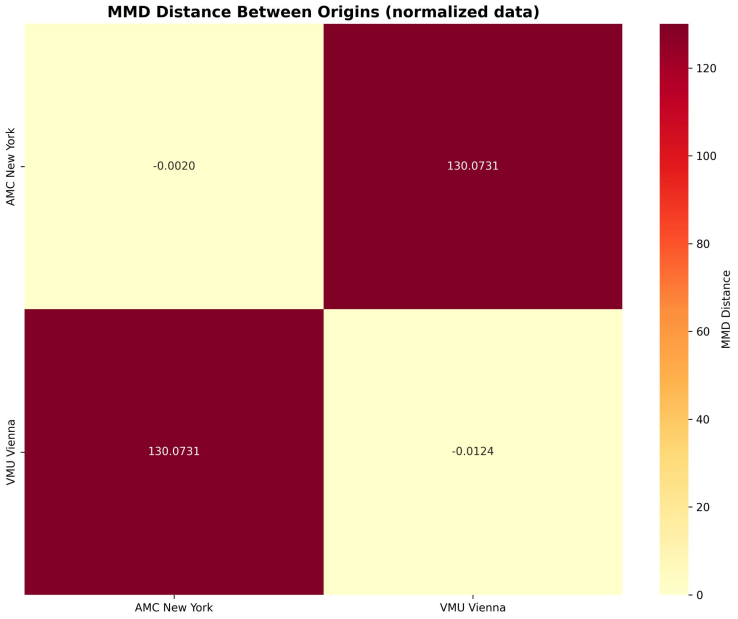


Lab Origin

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
6a	85	soft tissue sarcoma	AMC New York	canine	3DHistech Pannoramic Scan II	$0.25\frac{\mu m}{px}$
6b	15	soft tissue sarcoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25\frac{\mu m}{px}$

Lab Origin

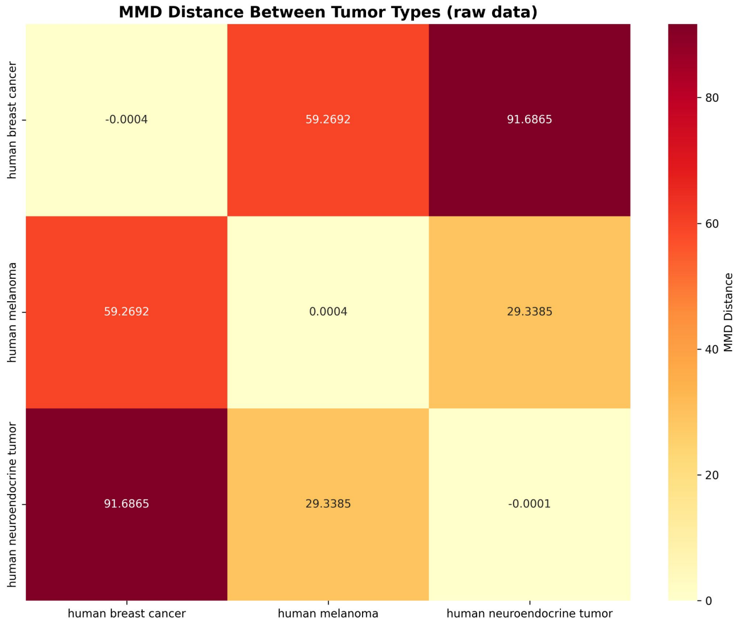

MMD Distance Between Origins (raw data)
AMC New York
VMU Vienna
-0.0024
144.5934
144.5934
-0.0182
MMD Distance
0
20
40
60
80
100
120
140

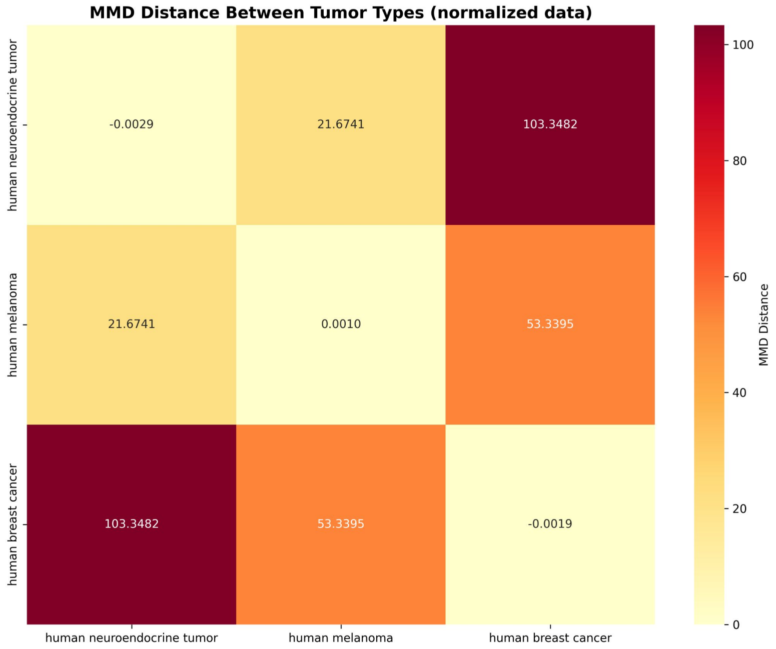

MMD Distance Between Origins (normalized data)
AMC New York
VMU Vienna
-0.0020
130.0731
130.0731
-0.0124
MMD Distance
0
20
40
60
80
100
120

Tumor Differences

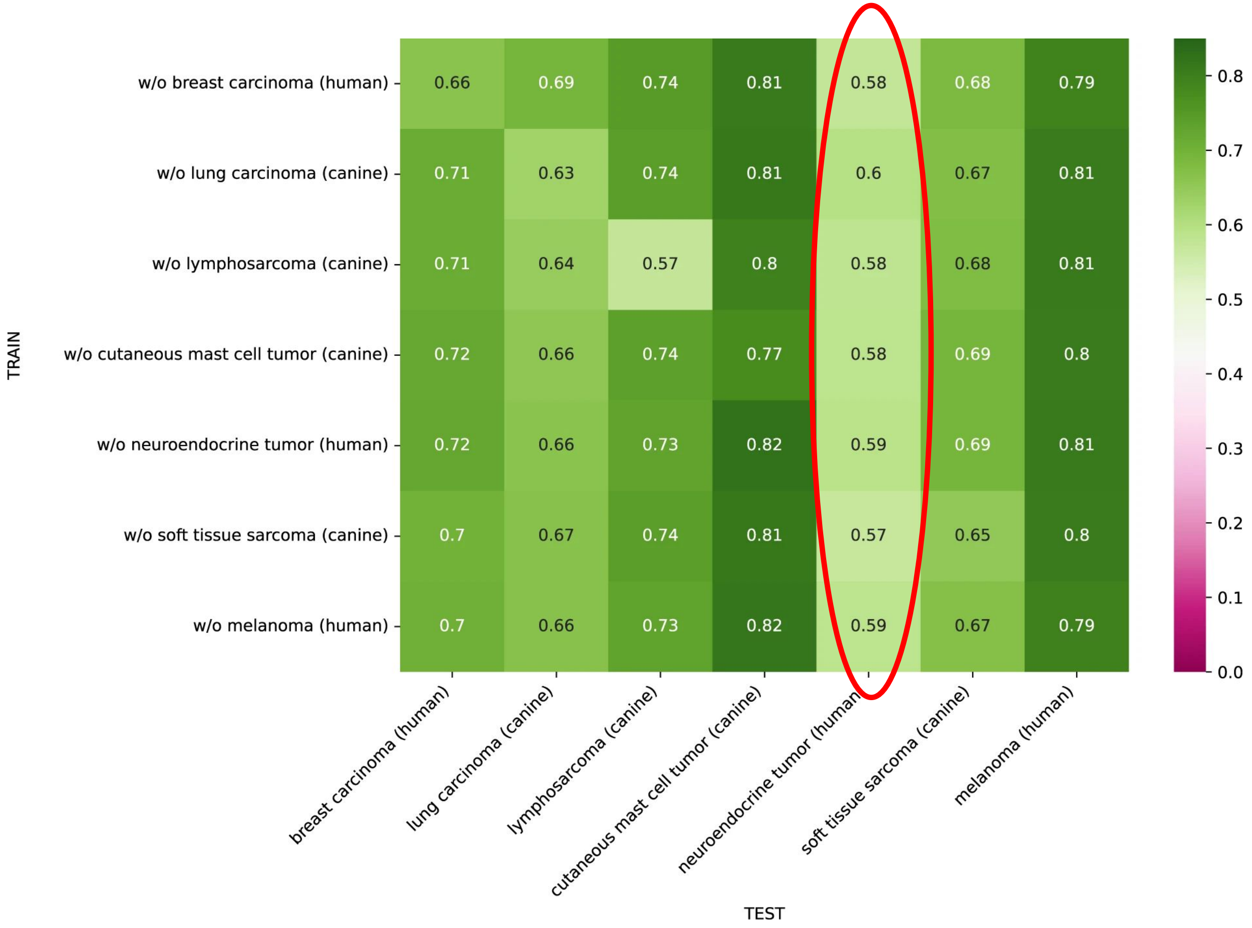
Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23\frac{\mu m}{px}$
5	55	neuroendocrine tumor	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23\frac{\mu m}{px}$
7	49	melanoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23\frac{\mu m}{px}$

Tumor Differences





Leave-One-Domain-Out Training from Paper



Summary

-Normalization reduced the MMD in similar Scanner

-Slightly reduced MMD within lab origin and some tumor type

Next Steps

-Domain shift similarity with CORAL

-Clustering within feature space (look at Silhouette Score)

Thank You